

MODULE 14

Urinary and Reproductive
Systems

Urinary System

Filtration of Blood: The kidneys filter out waste products, toxins, and excess substances from the bloodstream.

Regulation of Blood Volume and Pressure: By adjusting the amount of water excreted, the kidneys help control blood volume and, in turn, blood pressure.

Electrolyte Balance: The urinary system maintains the balance of electrolytes like sodium, potassium, and calcium. These electrolytes are crucial for muscle function, nerve transmission, and other bodily processes.

Acid-Base Balance: The kidneys help regulate the body's pH by excreting hydrogen ions and reabsorbing bicarbonate from urine.

Excretion of Waste: Waste products, like urea and creatinine, are removed from the body in the form of urine.

Hormone Production: The kidneys produce hormones such as erythropoietin, which stimulates red blood cell production, and renin, which helps regulate blood pressure.

Detoxification: The kidneys assist in the detoxification process by excreting metabolic by-products and drugs.